ROS2 Assignment report

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List of errors:

1) Don't put def main() inside the class. It does not have an indent.
2) If there are more than one publishers, don't use self.sub for all. Use self.sub1, self.sub2 etc.
3) Use different msg variables if both publishing and subscribing in the same node (causes shadowing).
4) Dont tick debug in hide rqt_graph coz it hid monitor node for some reason (dont know why yet)
5) Use separate functions for timer and subscription not the same function as args are different.
6) INITIALIZE ALL self. Variables in init
7) Use msg.header.stamp not just msg.header
8) Always add default value in declare_parameter
9) The launch file should be in the same folder as setup.py not further inside the package.
10) When updating odometry keep separate functions for timer and subscription.
11) It is rcl_interfaces.msg NOT rclpy_interfaces.msg (crying emoji)
12) **Single Invocation:** Combine all .msg, .srv, and .action relative file paths under one rosidl_generate_interfaces block.
13) Didnt understand reentrant_callback_group part
14) In mission_client, use sequential callbacks not asynchronous as ctrl+c should register as a command not shut down the whole node.
15) Create a separate topic for navigator_node(cmd_vel_nav) so that only one node publishes on cmd_vel.

How a service works:

Instead of message(msg), services and clients use srv files. Srv files have 2 parts, request and response. The client server sends the srv file containing the request to the server. The server receives the request. The server then returns the response based on the request.
It is different from a topic and action in the sense of the number of things present. A topic only has information. A service has a request and a response. An action has a goal, result and feedback.

Comments on assignment:

I struggled quite a bit on task 8 and 9 and a little bit on task 7. Also I forgot to take a screenshot of task 3. Sorry about that.